

James Shanley Pratt, MEng, AMIChemE

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Education

University of Newcastle

Master of Chemical Engineering (MEng, 2:1)

Key Technical Skills

AI / Data / Cloud:

- **Python** (2 years, data manipulation and gathering, APIs, web scraping)
- **Terraform** (2 years, infrastructure automation)
- **Google Cloud Platform** (2 years, Terraform provisioning)
- Predictive & **Machine Learning** Control for chemical processes (6-month research placement)
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Engineering & Process:

- **HAZOP & HAZID** (3rd year university project for methanol synthesis)
- **PFDs** & design using Visio and AutoCAD (3rd year university project and HVAC design engineer for Prowess Building Services)
- **Process control** (6 month paid research project and additional university modules)
- **Process modelling** of the Maillard reaction using python to determine how a change in condition change the reaction

Professional Experience

2022 – 2024

Data Engineer / Cloud Infrastructure Engineer (CIE) • Lloyds Bank (Placement via Kubrick Group Consultancy)

- Led cloud infrastructure initiatives using Terraform and Python to automate processes and improve efficiency across GCP workloads.
- Migrated data from Hadoop local storage to Google Cloud using CI/CD pipelines and Jenkins.
- Decommissioned Hadoop File System by implementing access restrictions to assess usage and requirements.
- Managed IAM permissions in Hadoop and GCP, ensuring least-privilege access.
- Updated and tested GCP systems (e.g., Dataproc), identifying bugs and applying fixes during rollout.
- Implemented guardrails to reduce errors and computational loads by enforcing correct inputs at workflow entry points.
- Performed root cause analysis of infrastructure errors, documenting and applying solutions.

- Built dashboards to monitor cloud performance and usage, identifying redundant files and saving thousands of pounds per month.
- Automated label identification and addition in Terraform files with Python, replacing manual processes.
- Worked within Agile and Kanban frameworks to track progress and deliver solutions.

2022

Data Engineer • Medical Aid Films (via Kubrick Group Consultancy)

- Gathered data from multiple social media platforms using APIs.
- Built and tested pipelines locally, producing BI reports with PowerBI and Tableau.
- Migrated local databases to Google Cloud for automated reporting.

2022

Data Engineer • Kubrick Group Consultancy

- Completed formal training in Python, SQL, BI, and visualisation tools (PowerBI, Tableau).
- Team project: Web-scraped quarterly business reports, cleaned data for ELT, performed sentiment analysis, and compared results with stock market values via APIs.
- Individual project: Web-scraped unstructured data and compiled into organised spreadsheets for improved visualisation and editing.

2021

Researcher / Report Writer • Newcastle University & Optimal Industrial Technologies

- Developed and analysed predictive control techniques to improve industrial process reliability.
- Designed Python simulations to optimise process efficiency and diagnose control issues.
- Simulations include python libraries **sci-kit, TensorFlow, NumPy, Pandas**
- Delivered weekly presentations to managers and produced a final report recommending suitable methods for process optimisation.

2016

Engineer • Prowess Building Services

- Produced initial and detail design stages for HVAC systems and plantrooms in AutoCAD for commercial and domestic usage
- Produced detailed flow PID diagrams with full safety systems
- Full pressure drop calculations, heat requirements, and equipment sizing for HVAC systems including material selection for pipework, pumps, duct work and heat exchangers

Achievements

- Built a Python-based automated label verification system to streamline compliance checks, reducing manual workloads and ensuring deployment compliance.

- Applied test-driven development to maintain code quality; presented the system to Lloyds Bank's Data & Machine Learning department.
- Created GCP migration dashboards that identified redundant test files, saving thousands of pounds per week in storage costs.
- Diagnosed and fixed quota bottlenecks in processing pipelines, preventing downstream errors.
- Created full methanol synthesis plant for 3rd year university project
- Paid research project with Optimal Industrial Technology LTD researching and evaluating Advanced predictive control techniques for reaction control

Additional Information

- Certifications & Documents: Full UK driving licence, British and Irish passports, EFR First Responder & Secondary Care (First Aid), PADI Divemaster